

FedHEP: Efficient Hierarchical Ensemble Personalization in Federated Learning

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Abstract

Federated Learning (FL) on resource-constrained edge devices faces three challenges: mitigating client drift under non-independent and identically distributed (non-IID) data, enabling personalization without dual-model memory overhead, and defending against Byzantine poisoning without penalizing honest specialized clients. We introduce **FedHEP** (*Federated Hierarchical Ensemble Personalization*), a single-backbone framework coordinating three representation tiers—global consensus, collaborative peer clusters, and private local heads. First, Active-Class Logit Masking (**ACLM**) with skew-calibrated loss weighting suppresses gradient interference from unobserved categories without duplicating feature extractors. Second, low-dimensional Random Projections enable privacy-preserving cluster discovery among peers. Third, Subspace-Constrained Cosine Filtering (**SCCF**) coupled with Temporal Trust Tracking (**TTT**) enhances Byzantine resilience by evaluating active-class alignment to reduce false-positive rejection of specialized clients. Experiments on CIFAR-100 and FEMNIST across five Dirichlet non-IID regimes, multi-attack Byzantine benchmarks, and 50-client scaling show that FedHEP improves local test accuracy and worst-decile tail fairness over evaluated baselines. Simulated on-device runtime and memory profiling on MobileNetV3 demonstrates single-model memory footprint and latency efficiency for resource-constrained edge deployment.

Keywords

Federated Learning, Multi-Scale Personalization, Byzantine Fault Tolerance, Statistical Heterogeneity, Tail Fairness

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1 Introduction

Federated Learning (FL) enables distributed edge clients to collaboratively train neural network models without sharing private local data [17, 24]. In practical deployments on resource-constrained embedded and mobile nodes, collaborative learning faces three key challenges spanning data heterogeneity, edge hardware limits, and adversarial attacks.

First, local datasets across edge nodes are non-independent and identically distributed (**non-IID**)—departing from independent and

identically distributed (**IID**) data—with severe label skew [15]. Under this heterogeneity, global methods like Federated Averaging (FedAvg) [24] and FedProx (Federated Proximal) [21] suffer from *client drift*, where conflicting updates degrade accuracy on specialized nodes [22].

Second, while personalized FL mitigates client drift, existing schemes incur high on-device memory costs. Dual-model architectures (e.g., Ditto [20]) maintain concurrent global and personalized networks per device, doubling parameters, backpropagation activations, and peak video random-access memory (VRAM). Split-head frameworks such as FedRep (Federated Representation Learning) [7] decouple a shared backbone from private classifier heads; however, isolated local heads do not collaborate with peers and are prone to overfitting when local samples are scarce.

Third, adversarial model poisoning and Byzantine failures pose serious threats in distributed networks [3, 10, 30]. Standard robust aggregators (such as Multi-Krum [3], coordinate-wise median, and trimmed mean) evaluate model updates in the full Euclidean parameter space. When data is non-IID, honest updates from specialized clients naturally point in different directions from the average update. As a result, standard Euclidean distance filters often mistakenly reject honest specialized clients while failing to filter subtle coordinated attacks.

To address these challenges, we observe that edge clients benefit from a multi-tier representation hierarchy: a global model captures features common across all clients, cluster-level models pool knowledge among peer devices with similar label distributions, and local heads provide private adaptation tailored to each device. To avoid the memory overhead of maintaining separate neural networks for each tier, we design a unified framework where all three tiers share a single feature backbone.

We introduce **FedHEP** (*Federated Hierarchical Ensemble Personalization*), an efficient framework designed for heterogeneous edge clients. Our key contributions are:

Single-Backbone Three-Tier Hierarchy: We design an architecture where a single shared feature extractor feeds three linear classifier head weight matrices spanning C classes for d -dimensional embeddings: a global Root head ($W_r \in \mathbb{R}^{d \times C}$), a cluster Parent head ($W_p \in \mathbb{R}^{d \times C}$), and a private Local head ($W_l \in \mathbb{R}^{d \times C}$), avoiding redundant backbones and minimizing edge memory.

ACLM and Skew-Calibrated Weighting: We introduce Active-Class Logit Masking (**ACLM**) to mask unobserved categories on cluster and local heads, suppressing gradient corruption from absent classes on personalized heads, while Bernstein polynomials balance each tier’s loss contribution based on local label skew.

Privacy-Preserving Peer Clustering: Clients discover collaborative peer clusters using low-dimensional Random Projections (RP)



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grounded in the Johnson-Lindenstrauss (JL) lemma [16, 29], grouping clients by functional update similarity without transmitting raw local parameters or label counts.

Layered Subspace Byzantine Defense: We develop a defense combining Subspace-Constrained Cosine Filtering (SCCF), measuring update alignment on observed classes, adaptive backbone update norm-bounding, and multi-round Temporal Trust Tracking (TTT).

Empirical Evaluation: We evaluate FedHEP on edge-budgeted Federated Extended MNIST (FEMNIST) and CIFAR-100 across five Dirichlet non-IID regimes with concentration parameter $\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$ ($\alpha = \infty$ denotes balanced IID) and 50-client scaling under partial participation. We compare against 6 representative baselines across 4 attack vectors, and profile simulated runtime on MobileNetV3 to evaluate edge efficiency.

2 Related Work

Personalized Federated Learning. To combat non-IID client drift, personalized federated learning departs from monolithic consensus. Consensus regularization like FedProx [21] and SCAFFOLD (Stochastic Controlled Averaging) [18] stabilizes optimization via proximal terms or control variates, yet restricts clients to a single consensus model. Meta-learning such as Per-FedAvg (Personalized FedAvg) [9] adapts a meta-model, but incurs costly Hessian computations. Dual-model architectures (e.g., APFL [Adaptive Personalized FL] [8] and Ditto [20]) decouple global and personalized objectives via two concurrent client networks, doubling parameter storage and backpropagation memory.

Head-decoupling formulations (e.g., FedPer [Personalization layers] [1], FedRep [Representation learning] [7], FedBABU [Body-Aggregation & Bone-Updating] [26], FedRoD [Robust Decoupled FL] [6], and FedALA [Adaptive Local Aggregation] [31]) split classifier heads from a shared backbone. However, localized heads remain isolated from peers. As shown in recent work [6] and our benchmarks (Section 4), scarce local data causes isolated heads to overfit under severe skew; conversely, on balanced IID data, they fail to learn the full label space without classifier aggregation.

Byzantine-Robust Aggregation under Heterogeneity. Federated learning is vulnerable to adversarial poisoning, including label flipping, sign flipping, and gradient ascent [2, 3, 10, 28, 30]. Classical robust aggregators—coordinate-wise median, trimmed mean [30], and Multi-Krum [3]—assume honest client updates concentrate around a unimodal mean. Under non-IID skew, honest updates disperse angularly, causing standard Euclidean filters to misclassify specialized nodes as malicious. While proxy schemes like FLTrust [5] avoid this via central data, collecting edge samples centrally violates decentralized privacy.

Clustered Learning and Dimensionality Reduction. Clustered FL groups clients into discrete coalitions. Clustered FL (CFL) [27] partitions clients by stationary gradient angles, while IFCA (Iterative Federated Clustering Algorithm) [11] clusters via loss minimization. However, existing methods either transmit raw gradients—risking class privacy—or lack defenses against Byzantine cluster

manipulation. Meanwhile, Johnson-Lindenstrauss (JL) random projection sketching [16, 29] preserves pairwise metrics, but has not been integrated into hierarchical multi-tier personalization pipelines.

Unlike methods addressing personalization, Byzantine robustness, or efficiency in isolation, FedHEP unifies them into a single-backbone framework. Combining Active-Class Logit Masking (ACLM) with sketching for peer discovery and subspace-constrained filtering (SCCF), FedHEP demonstrates high personalization accuracy, outlier filtering under skew, and single-model edge memory efficiency.

3 The FedHEP Methodology

High-Level Workflow. Figure 1 and Algorithm 1 summarize the overall architecture and execution workflow of FedHEP. On-device, each client maintains a single shared convolutional backbone Φ_θ feeding three linear heads (W_r, W_p, W_l), applying Active-Class Logit Masking (ACLM) and skew-calibrated loss weighting to suppress absent-class interference. For peer discovery, clients compress updates into random sketches s_i . On the server, Root updates and backbones are aggregated via Subspace-Constrained Cosine Filtering (SCCF), adaptive norm clipping, and Temporal Trust Tracking (TTT).

Problem Formulation. We consider a distributed federated network of N heterogeneous edge clients $\mathcal{N} = \{1, \dots, N\}$. Each client i possesses a private local training dataset $\mathcal{D}_i = \{(x_j, y_j)\}_{j=1}^{n_i}$ sampled from distribution $\mathcal{P}_i(x, y)$ over input space $\mathcal{X}$ and label space $\mathcal{Y} = \{0, 1, \dots, C - 1\}$. We denote the subset of classes present on client i by $\mathcal{Y}_i = \{y \in \mathcal{Y} \mid \exists (x, y) \in \mathcal{D}_i\}$. Under non-IID Dirichlet label partitioning, edge clients typically observe only a fraction of the total classes ($|\mathcal{Y}_i| \ll C$).

Threat Model. We consider a Byzantine adversary controlling up to a fraction $q < 0.5$ of clients $\mathcal{M} \subset \mathcal{N}$ executing coordinated attacks [2, 10, 28]: (1) *Label Flipping*: malicious nodes map training labels symmetrically via $y \rightarrow C - 1 - y$; (2) *Sign Flipping*: updates are inverted and scaled via $\Delta \mathbf{w} \rightarrow -\gamma \Delta \mathbf{w}$ (with scaling factor $\gamma > 0$); (3) *Gradient Ascent*: implemented as an amplified update reversal $\Delta \mathbf{w} \rightarrow -5 \Delta \mathbf{w}$ (versus $\gamma = 1.5$ for Sign Flipping), which pushes the aggregate to ascend the training loss; and (4) *Gaussian Noise*: updates are replaced with random Gaussian vectors $\mathcal{N}_{\text{Gauss}}(0, \sigma^2 \mathbf{I})$. The central server maintains no clean proxy validation dataset and cannot inspect private client data.

3.1 Hierarchical Representation Architecture

Architecture Overview and Intuition. Personalized federated learning on edge devices faces a fundamental tension between model expressiveness and on-device resource constraints. Conventional dual-model formulations (e.g., Ditto [20]) maintain two full neural network instances per client—one global consensus model and one personalized model—which roughly doubles parameter storage, backpropagation activations, and peak VRAM. Conversely, decoupled-head approaches (e.g., FedRep [7]) train a single backbone alongside an isolated local head, but omit peer collaboration; when local training data is scarce under severe label skew, these isolated heads are prone to overfitting on narrow class subsets. FedHEP addresses this trade-off by introducing a tripartite classifier hierarchy built

on top of a single shared convolutional feature extractor $\Phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$. Rather than duplicating backbones or isolating clients, FedHEP structures classification capacity into three complementary tiers: (1) a globally aggregated Root head (W_r) capturing invariant representations common across all N clients; (2) a collaborative Parent head (W_p) synchronized among peer clients within cluster $\mathcal{C}_k$ sharing overlapping label distributions; and (3) a private Local head (W_l) optimized on-device without communication. During on-device training, Active-Class Logit Masking (ACLM) shields the shared backbone from gradient interference caused by unobserved classes, while skew-calibrated loss weighting dynamically shifts capacity across tiers based on empirical label entropy. During inference, predictions are formed through a temperature-calibrated linear ensemble of the three heads, achieving strong personalization accuracy and tail fairness within the footprint of a single edge model.

Three-Tier Classifier Heads on a Single Backbone. Specifically, each client maintains a single convolutional feature extractor $\Phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ parameterized by weights θ , where d denotes the feature representation dimension (e.g., $d = 512$). Branched directly from the terminal feature representation $\Phi_\theta(x)$ are three linear classification heads $W_h \in \mathbb{R}^{d \times C}$ indexed by tier $h \in \{r, p, l\}$, where C is the total class count: the Root head W_r (aggregated globally across $\mathcal{N}$ via Byzantine-resilient consensus), the Parent head W_p (synchronized within peer cluster $\mathcal{C}_{k(i)}$), and the Local head W_l (trained locally on-device).

During on-device inference, predictions are computed by an anchored linear combination of temperature-calibrated logits [12]:

$$\mathbf{z}_{\text{ens}} = w_l \frac{\mathbf{z}_i^{(l)}}{T_l} + w_p \frac{\mathbf{z}_i^{(p)}}{T_p} + w_r \frac{\mathbf{z}_i^{(r)}}{T_r}, \quad \hat{y} = \arg \max_{c \in \mathcal{Y}} z_{\text{ens},c}, \quad (1)$$

where $\mathbf{z}_i^{(h)} = W_h^\top \Phi_\theta(x) \in \mathbb{R}^C$ is the logit vector produced by tier $h \in \{r, p, l\}$, $z_{\text{ens},c}$ denotes the c -th coordinate of composite logit $\mathbf{z}_{\text{ens}}$, $\hat{y} \in \mathcal{Y}$ is the predicted class label, and $T_h > 0$ are fixed head-specific temperature scalars that calibrate prediction confidence across heads (empirically set to $T_r = 1.0$, $T_p = 0.8$, and $T_l = 0.5$). The coefficients $w_h \geq 0$ (satisfying $\sum_h w_h = 1$) denote inference ensemble weights, which are initialized from the skew-calibrated training weights α_h derived below and refined online. At evaluation, FedHEP restricts the argmax to the client's observed classes $\mathcal{Y}_i$ (closed-set inference) in the main benchmarks; baselines in those benchmarks use unrestricted argmax (see the open-set results in Section 4.2).

Active-Class Logit Masking (ACLM). In standard cross-entropy training, computing the softmax denominator over all C classes forces the network to assign non-zero probabilities to unobserved classes. When a client only possesses a subset of classes $\mathcal{Y}_i \subset \mathcal{Y}$, evaluating standard cross-entropy penalizes these missing categories, driving their logits toward $-\infty$ and injecting destructive negative gradient updates that degrade the shared backbone Φ_θ [25].

To mitigate this gradient interference, FedHEP applies Active-Class Logit Masking (ACLM) to the Parent and Local heads. For client i with observed class set $\mathcal{Y}_i$, the masked logit coordinate for

class $c \in \{0, \dots, C-1\}$ is defined as:

$$\tilde{z}_{i,c} = \begin{cases} z_{i,c}, & \text{if } c \in \mathcal{Y}_i, \\ -\infty, & \text{if } c \notin \mathcal{Y}_i. \end{cases} \quad (2)$$

Mathematical Derivation of Zero-Gradient Shielding: Let (x, y) be a training sample on client i , where the ground-truth label satisfies $y \in \mathcal{Y}_i$. The masked softmax probability for class c is:

$$\tilde{p}_c = \frac{\exp(\tilde{z}_{i,c})}{\sum_{j \in \mathcal{Y}} \exp(\tilde{z}_{i,j})} = \frac{\exp(\tilde{z}_{i,c})}{\sum_{j \in \mathcal{Y}_i} \exp(z_{i,j}) + \sum_{j \notin \mathcal{Y}_i} \exp(-\infty)}. \quad (3)$$

For any unobserved category $c \notin \mathcal{Y}_i$, since $\lim_{z \rightarrow -\infty} \exp(z) = 0$, the numerator is 0, which gives $\tilde{p}_c = 0$. The gradient of standard cross-entropy (CE) loss $\mathcal{L}_{\text{CE}} = -\ln \tilde{p}_y$ with respect to logit coordinate $z_{i,c}$ is:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} = \tilde{p}_c - \mathbb{I}(y = c), \quad (4)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function ($\mathbb{I} = 1$ if the condition holds, else 0). Since true label $y \in \mathcal{Y}_i$, for all unobserved classes $c \notin \mathcal{Y}_i$ we have $\mathbb{I}(y = c) = 0$. Combined with $\tilde{p}_c = 0$, Equation (4) yields:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} = 0 - 0 = 0, \quad \forall c \notin \mathcal{Y}_i. \quad (5)$$

By the multivariate chain rule, the backpropagated gradient from classifier head $W \in \mathbb{R}^{d \times C}$ into the shared feature extractor $\Phi_\theta(x)$ is $\nabla_{\Phi_\theta(x)} \mathcal{L}_{\text{CE}} = \sum_{c \in \mathcal{Y}} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} \mathbf{w}_c = \sum_{c \in \mathcal{Y}_i} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} \mathbf{w}_c + \mathbf{0}$, where $\mathbf{w}_c \in \mathbb{R}^d$ denotes the column of head weight matrix W corresponding to class c . Consequently, unobserved categories contribute exactly zero gradient to the shared feature extractor from the Parent and Local heads, shielding shared representations from label-absence penalties.

Skew-Calibrated Loss Weighting. To balance the three classification tiers, each client dynamically weights its loss terms based on its empirical label distribution. Client i first computes its label frequency vector $\mathbf{p}_i = [p_{i,0}, \dots, p_{i,C-1}]^\top$, where $p_{i,c} = \frac{|\{(x,y) \in \mathcal{D}_i : y=c\}|}{|\mathcal{D}_i|}$, and empirical Shannon entropy $H(\mathbf{p}_i) = -\sum_{c \in \mathcal{Y}_i} p_{i,c} \ln p_{i,c}$. The effective class count is quantified by the order-1 Hill number (perplexity) $\exp(H(\mathbf{p}_i))$. Normalizing this effective diversity over $[1, C]$ defines the skew ratio:

$$r_{\text{skew}}^{(i)} = \frac{\exp(H(\mathbf{p}_i)) - 1}{C - 1} \in [0, 1]. \quad (6)$$

When local data is concentrated in a single class (extreme skew), $H(\mathbf{p}_i) = 0$ yielding $r_{\text{skew}}^{(i)} = 0$; conversely, under a balanced uniform distribution across all C classes (IID), $H(\mathbf{p}_i) = \ln C$ yielding $r_{\text{skew}}^{(i)} = 1$.

To ensure specialized clients retain an anchor to global consensus, we define baseline floor $a_i = \max(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C})$, with K clusters. Using degree-2 Bernstein basis polynomials [23], which form a partition of unity over $[0, 1]$, tier weights are unnormalized as:

$$\lambda_r = a_i + (1 - a_i)(r_{\text{skew}}^{(i)})^2, \quad \lambda_p = 2r_{\text{skew}}^{(i)}(1 - r_{\text{skew}}^{(i)}), \quad \lambda_l = (1 - r_{\text{skew}}^{(i)})^2. \quad (7)$$

These polynomials smoothly govern capacity allocation across regimes: as $r_{\text{skew}}^{(i)} \rightarrow 1$, capacity concentrates on the Root head (λ_r); as

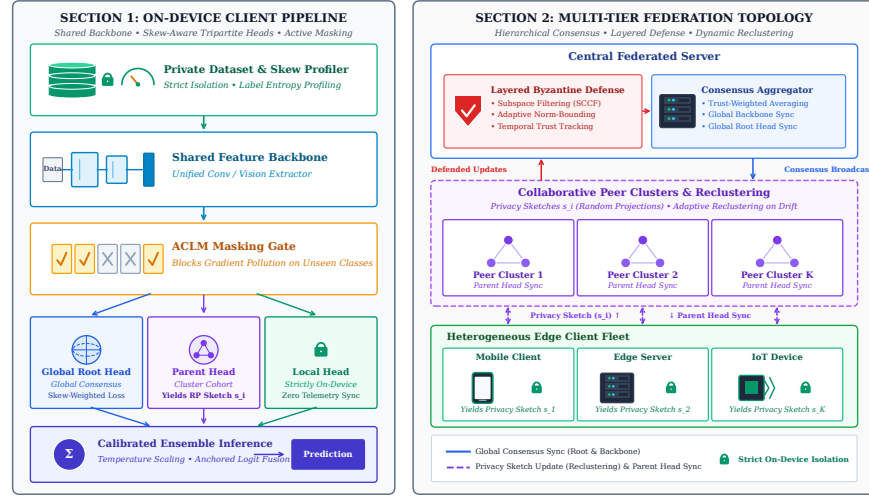


Figure 1: Overview of FedHEP Architecture. (Left) On-device client with a single backbone, tripartite head ensemble, and Active-Class Logit Masking (ACLM). (Right) Server-side sketch clustering and layered subspace defense (SCCF + TTT).

$r_{\text{skew}}^{(i)} \rightarrow 0$, private Local capacity (λ_l) dominates while a_i anchors the shared backbone; and under intermediate skew ($r_{\text{skew}}^{(i)} \approx 0.5$), collaborative Parent weighting (λ_p) peaks. Normalized coefficients $\alpha_h = \lambda_h / \sum_j \lambda_j$ form the composite objective:

$$\mathcal{L}_{\text{total}} = \alpha_r \mathcal{L}_r(W_r; \theta) + \alpha_p \mathcal{L}_p^{\text{mask}}(W_p; \theta) + \alpha_l \mathcal{L}_l^{\text{mask}}(W_l; \theta), \quad (8)$$

where $\mathcal{L}_r$ is unmasked cross-entropy, and $\mathcal{L}_p^{\text{mask}}$, $\mathcal{L}_l^{\text{mask}}$ denote cross-entropy evaluated over ACLM-masked logits.

Privacy-Preserving Peer Clustering via Random Projections (RP). To group clients into collaborative peer clusters without sharing raw model weights or label histograms, clients compress their Parent head updates $\Delta W_{p,i} \in \mathbb{R}^{d \times C}$ into compact sketches. Let $\text{vec}(\Delta W_{p,i}) \in \mathbb{R}^D$ denote the vectorized parameter update ($D = d \cdot C$, e.g., $512 \times 100 = 51,200$). Clients project this vector using a shared Gaussian matrix $\mathbf{R} \in \mathbb{R}^{d_{\text{proj}} \times D}$ ($\mathbf{R}_{ab} \sim \mathcal{N}(0, 1/d_{\text{proj}})$) with sketch dimension $d_{\text{proj}} = 256$), followed by L_2 normalization:

$$\mathbf{s}_i = \frac{\mathbf{R} \cdot \text{vec}(\Delta W_{p,i})}{\|\mathbf{R} \cdot \text{vec}(\Delta W_{p,i})\|_2} \in \mathbb{R}^{d_{\text{proj}}}. \quad (9)$$

By the Johnson-Lindenstrauss lemma [16, 29], pairwise Euclidean distances between normalized sketches approximate the cosine distance between original updates within metric distortion tolerance $\epsilon \in (0, 1)$ with high probability whenever $d_{\text{proj}} \geq \mathcal{O}(\epsilon^{-2} \log N)$. Because $d_{\text{proj}} = 256 \ll D \approx 5 \times 10^4$, inverting the sketch to reconstruct original weights is mathematically underdetermined, preserving edge privacy while enabling the server to partition clients into K synergistic clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ using spherical k -means.

3.2 Layered Subspace Byzantine Defense

Defense Overview and Intuition. Defending federated learning against Byzantine poisoning under severe statistical heterogeneity presents a fundamental dilemma. Classical robust aggregators

(such as Multi-Krum [3], coordinate-wise median, and geometric median [30]) rely on Euclidean distance metrics in parameter space $\mathbb{R}$ under the premise that benign client updates concentrate tightly around a single consensus mean. However, as illustrated in Figure 2(a), this premise breaks down in non-IID edge networks. When clients observe only a fraction of classes ($|y_i| \ll C$), honest specialized updates point into sparse, mutually orthogonal coordinate subspaces. Consequently, the Euclidean distance between an honest specialized client's update and the global average update is often just as large as that of a malicious poisoning vector. Standard Euclidean outlier detectors therefore misclassify legitimate specialized clients as Byzantine attackers, triggering false-positive rejection that discards informative local domain updates and damages tail fairness.

To address this robustness-heterogeneity dilemma, FedHEP implements a three-tier defense pipeline on the server that decouples classifier head filtering from shared backbone aggregation: (1) *Subspace-Constrained Cosine Filtering (SCCF)* evaluates directional update alignment within each client's observed class subspace $\mathcal{S}_i$ (Figure 2(b)), where honest non-IID updates exhibit positive alignment with consensus ($\cos > 0$) while Byzantine vectors remain separated and down-weighted; (2) *Adaptive Backbone Norm-Bounding* constrains backbone update magnitudes using a dynamic threshold anchored to the first quartile (Q_1) of received client update norms, limiting adversarial manipulation of clipping bounds; and (3) *Temporal Trust Tracking (TTT)* maintains exponential moving average credibility scores across communication rounds, mitigating intermittent or stealthy adversaries and permanently isolating persistent attackers once cumulative trust falls below threshold τ_{iso} .

Subspace-Constrained Cosine Filtering (SCCF). To reduce false-positive outlier filtering while maintaining Byzantine resilience, the server evaluates uploaded Root head updates $\Delta \mathbf{w}_i = \text{vec}(\Delta W_{r,i}) \in$

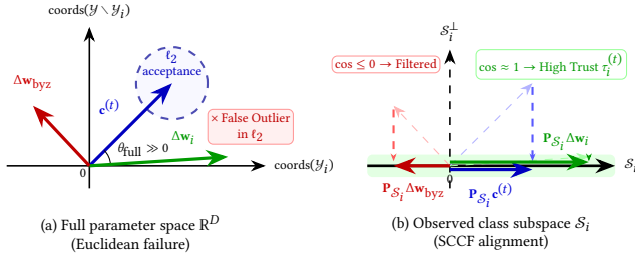


Figure 2: Geometric Intuition of Subspace-Constrained Cosine Filtering (SCCF). (a) In full parameter space $\mathbb{R}^D$, an honest client with non-IID label skew ($\mathcal{Y}_i \subset \mathcal{Y}$) updates primarily along active coordinates $\text{coords}(\mathcal{Y}_i)$, creating large Euclidean distance to consensus $\mathbf{c}^{(t)}$ and triggering false rejection. (b) Projecting via $\mathbf{P}_{S_i}$ onto active subspace S_i removes unobserved dimensions ($S_i^\perp$): honest updates align with consensus ($\cos \approx 1$, high trust $\tau_i^{(t)}$), while Byzantine attacks remain separated.

$\mathbb{R}^D$ within observed class subspace S_i . For each observed class $c \in \mathcal{Y}_i$, its d classifier weights in the flattened vector are indexed by $\text{coords}(c) = \{c \cdot d, \dots, (c+1)d - 1\}$. We denote client i 's active coordinate set by $\text{coords}(\mathcal{Y}_i) = \bigcup_{c \in \mathcal{Y}_i} \text{coords}(c)$, and define active subspace $S_i = \text{span}\{\mathbf{e}_j : j \in \text{coords}(\mathcal{Y}_i)\} \subset \mathbb{R}^D$, where $\mathbf{e}_j \in \mathbb{R}^D$ denotes the j -th standard canonical basis vector.

Let $\mathbf{P}_{S_i} \in \mathbb{R}^{D \times D}$ denote the diagonal coordinate projection operator:

$$[\mathbf{P}_{S_i}]_{jj} = \begin{cases} 1, & \text{if } j \in \text{coords}(\mathcal{Y}_i), \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

The server computes the normalized consensus direction vector $\mathbf{c}^{(t)}$ across participating clients $\mathcal{N}$:

$$\mathbf{c}^{(t)} = \frac{\sum_{j \in \mathcal{N}} \tilde{\mathbf{w}}_j}{\|\sum_{j \in \mathcal{N}} \tilde{\mathbf{w}}_j\|_2 + \epsilon_0}, \quad \text{where } \tilde{\mathbf{w}}_j = \frac{\mathbf{P}_{S_j} \Delta \mathbf{w}_j}{\|\mathbf{P}_{S_j} \Delta \mathbf{w}_j\|_2 + \epsilon_0}, \quad (11)$$

and $\epsilon_0 = 10^{-10}$ ensures numerical stability. Client i 's instantaneous trust score $\tau_i^{(t)}$ is computed via temperature-scaled softmax over active subspace cosine alignment across participating clients:

$$\tau_i^{(t)} = \frac{\exp(\cos(\mathbf{P}_{S_i} \Delta \mathbf{w}_i, \mathbf{P}_{S_i} \mathbf{c}^{(t)}) / T_{\text{def}})}{\sum_{j \in \mathcal{N}} \exp(\cos(\mathbf{P}_{S_j} \Delta \mathbf{w}_j, \mathbf{P}_{S_j} \mathbf{c}^{(t)}) / T_{\text{def}})}, \quad (12)$$

where $\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}$ is the cosine similarity and $T_{\text{def}} > 0$ is the defense temperature parameter. By restricting cosine similarity checks to observed active coordinates $\mathbf{P}_{S_i}$, SCCF isolates legitimate domain specialization from adversarial corruption. The server aggregates Root head updates using these trust weights: $\Delta \mathbf{w}_{\text{global}}^{(t)} = \sum_{i \in \mathcal{N}} \tau_i^{(t)} \mathbf{P}_{S_i} \Delta \mathbf{w}_i$.

Adaptive Backbone Norm-Bounding. To counter unbounded scaling attacks and gradient-ascent inflation on shared representations, backbone updates $\Delta \theta_i \in \mathbb{R}^{d_\theta}$ (d_θ backbone parameters) are constrained by an adaptive clipping threshold $\tau^{(t)}$:

$$\tilde{\Delta \theta}_i = \Delta \theta_i \cdot \min\left(1, \frac{\tau^{(t)}}{\|\Delta \theta_i\|_2}\right). \quad (13)$$

Rather than relying on a static, manually tuned clipping bound that risks either stifling honest early-round learning or admitting

Algorithm 1 FedHEP Client Training and Server Aggregation

Require: Clients $\mathcal{N}$, rounds R , clusters K , projection matrix $\mathbf{R} \in \mathbb{R}^{d_{\text{proj}} \times D}$

Ensure: Personalized models $(\Phi_\theta, W_r, W_p, W_l)$ for all $i \in \mathcal{N}$

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1: Server initializes  $\theta^{(0)}$ ,  $W_r^{(0)}$ , cluster heads  $\{W_{p,k}^{(0)}\}_{k=1}^K$ , trust  $T_i^{(0)} \leftarrow 1.0$ 
2: for each communication round  $t = 0, \dots, R - 1$  do
3:   Server broadcasts  $(\theta^{(t)}, W_r^{(t)})$  and cluster heads  $W_{p,k(i)}^{(t)}$ 
4:   for each client  $i \in \mathcal{N}$  in parallel do ▷ Client Local Training
5:     Compute skew ratio  $r_{\text{skew}}^{(i)}$  and weights  $(\alpha_r, \alpha_p, \alpha_l)$  via Eqs. (6)–(7)
6:     Train  $(\theta_i, W_{r,i}, W_{p,i}, W_{l,i})$  on  $\mathcal{D}_i$  minimizing  $\mathcal{L}_{\text{total}}$  with ACLM
7:     Compute deltas  $\Delta \theta_i, \Delta W_{r,i}, \Delta W_{p,i}$  and sketch  $\mathbf{s}_i$  via Eq. (9)
8:     Transmit updates  $(\Delta \theta_i, \Delta W_{r,i}, \Delta W_{p,i}, \mathbf{s}_i)$  to server
9:   end for
10:  Server updates clusters  $\{\mathcal{C}_k\}_{k=1}^K \leftarrow \text{SphericalKMeans}(\{\mathbf{s}_i\}, K)$ 
11:  Compute consensus  $\mathbf{c}^{(t)}$  (Eq. (11)) and clip bound  $\tau^{(t)}$  (Eq. (14))
12:  for each client  $i \in \mathcal{N}$  do ▷ Server Subspace Defense
13:    Compute trust score  $\tau_i^{(t)}$  via SCCF (Eq. (12)) and update  $T_i^{(t)}$  (Eq. (15))
14:    Clip backbone update:  $\tilde{\Delta \theta}_i \leftarrow \Delta \theta_i \cdot \min(1, \tau^{(t)} / \|\Delta \theta_i\|_2)$  (Eq. (13))
15:  end for
16:  Identify trusted cohort  $\mathcal{A}^{(t)} \leftarrow \{i \in \mathcal{N} \mid T_i^{(t)} \geq \tau_{\text{iso}}\}$ 
17:   $\theta^{(t+1)} \leftarrow \theta^{(t)} + \sum_{i \in \mathcal{A}^{(t)}} \tau_i^{(t)} \tilde{\Delta \theta}_i$ ;  $W_r^{(t+1)} \leftarrow W_r^{(t)} + \sum_{i \in \mathcal{A}^{(t)}} \tau_i^{(t)} \mathbf{P}_{S_i} \Delta W_{r,i}$ 
18:  Synchronize cluster heads  $\forall k: W_{p,k}^{(t+1)} \leftarrow W_{p,k}^{(t)} + \frac{1}{|\mathcal{C}_k \cap \mathcal{A}^{(t)}|} \sum_{i \in \mathcal{C}_k \cap \mathcal{A}^{(t)}} \Delta W_{p,i}$ 
19: end for
```

late-stage poisoning, the server sets $\tau^{(t)}$ dynamically each round:

$$\tau^{(t)} = k_{\text{clip}} \cdot Q_1 \left(\left\{ \|\Delta \theta_j^{(t)}\|_2 \right\}_{j \in \mathcal{N}} \right), \quad (14)$$

where Q_1 denotes the first quartile (25th percentile) norm across participating client updates, and $k_{\text{clip}} = 3.0$ is a robustness scaling factor. Anchoring to Q_1 ensures that adversaries controlling fraction $q < 0.25$ cannot artificially inflate the clipping threshold with oversized updates. Furthermore, as honest gradient norms naturally contract during model convergence, $\tau^{(t)}$ adaptively tightens, providing continuous defense against late-round drift. Clipped backbone updates are aggregated via trust-weighted consensus: $\Delta \theta_{\text{global}}^{(t)} = \sum_{i \in \mathcal{N}} \tau_i^{(t)} \tilde{\Delta \theta}_i$.

Temporal Trust Tracking (TTT). To suppress intermittent or stealthy adversaries that mimic benign updates in early rounds to build credibility before poisoning the global model, the server tracks an exponential moving average trust score across rounds:

$$T_i^{(t)} = \beta T_i^{(t-1)} + (1 - \beta) \cdot \min\left(1.0, |\mathcal{C}_{k(i)}| \cdot \tau_i^{(t)}\right), \quad (15)$$

where $T_i^{(t)} \in [0, 1]$ is client i 's cumulative trust score at round t (initialized to $T_i^{(0)} = 1.0$), $\beta \in [0, 1]$ is momentum ($\beta = 0.8$), $\mathcal{C}_{k(i)}$ is the peer cluster assigned to client i , and cluster size $|\mathcal{C}_{k(i)}|$ rescales instantaneous trust $\tau_i^{(t)}$ relative to uniform cluster baseline. Clients whose cumulative trust decays below isolation threshold τ_{iso} ($\tau_{\text{iso}} = 0.2$) are permanently isolated from aggregation, suppressing persistent Byzantine drift over multi-round training horizons.

4 Experimental Evaluation

4.1 Experimental Setup

We evaluate FedHEP on fine-grained vision benchmarks (Federated Extended MNIST [FEMNIST] [4] and CIFAR-100 [19]) partitioned under an edge-budgeted subsampling protocol to mirror on-device sample scarcity. Table 1 summarizes benchmark settings,

Table 1: Experimental Suite and Benchmark Parameters.

Component	Specification / Hyperparameters
Vision Benchmarks	FEMNIST [4] ($C = 62, N = 15, 18k$ samples) CIFAR-100 [19] ($C = 100, N = 15/50, 13k$ samples)
Data Heterogeneity	Dirichlet $\text{Dir}(\alpha)$ skew [15]: IID ($\alpha = \infty$), Mild ($\alpha = 1.0$), Mod ($\alpha = 0.5$), Sev ($\alpha = 0.1$), Ext ($\alpha = 0.05$)
Model Backbones	ResNet-9 [13] (1.65M params), MobileNetV3-Small [14] (3.01M params with 3 heads)
FL Baselines	<i>Consensus</i> : FedAvg [24], FedProx [21] ($\mu = 0.01$), SCAFFOLD [18]; <i>Personalized</i> : Ditto [20] ($\lambda = 0.05$), FedRep [7]; <i>Byzantine-Robust</i> : Multi-Krum [3]
Threat Models ($q \leq 0.4$)	Label Flip, Sign Flip, Gradient Ascent, Gaussian Noise [2, 10, 28]
Training Hyperparameters	Local epochs $E = 3$, Batch size $B \in \{32, 64\}$, Stochastic Gradient Descent (SGD) with momentum 0.9, learning rate $\eta \in [0.02, 0.05]$, weight decay 5×10^{-4}
Communication Budget	Rounds $R \in \{25, 30, 40\}$, 3 seeds {42, 123, 7}
Evaluation Metrics	Local Test Accuracy (%), Bottom-10% Tail Fairness [22] (%), Peak Video RAM (VRAM in MB), Latency (ms)

Dirichlet heterogeneity regimes [15], neural backbones (ResNet-9 [13] and MobileNetV3-Small [14]), threat models [2, 10, 28], and edge training configurations.

We compare FedHEP against six representative baselines spanning three paradigms: (1) *Global Consensus FL*: FedAvg [24], FedProx [21] (proximal weight $\mu = 0.01$), and SCAFFOLD [18]; (2) *Personalized FL*: Ditto [20] (personalization penalty $\lambda = 0.05$) and FedRep [7]; and (3) *Byzantine-Robust Aggregation*: Multi-Krum [3]. Our threat models cover four canonical attack vectors [2, 10, 28]: Label Flipping, Sign Flipping, Gradient Ascent, and Gaussian Noise Injection, across attacker ratios $q \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$.

All benchmark runs report mean and standard deviation across three random seeds ({42, 123, 7}). Performance is evaluated via Local Test Accuracy (mean Top-1 accuracy evaluated on each client’s private test partition), while worst-decile tail fairness is quantified via the Bottom-10% metric [22] (mean accuracy of the lowest-performing decile of edge nodes). The complete implementation and reproduction scripts are provided in the supplementary material (and will be released publicly upon publication).

4.2 Numerical Results

Personalization Performance under Statistical Heterogeneity. Table 2 and Figures 3–4 report test accuracy and tail fairness on FEMNIST and CIFAR-100 across five Dirichlet skew regimes, from balanced IID ($\alpha = \infty$) to extreme non-IID ($\alpha = 0.05$). In benign IID settings ($\alpha = \infty$), global consensus models (FedAvg, FedProx, SCAFFOLD) achieve solid accuracy ($\approx 84\%$ on FEMNIST, $\approx 41\%$

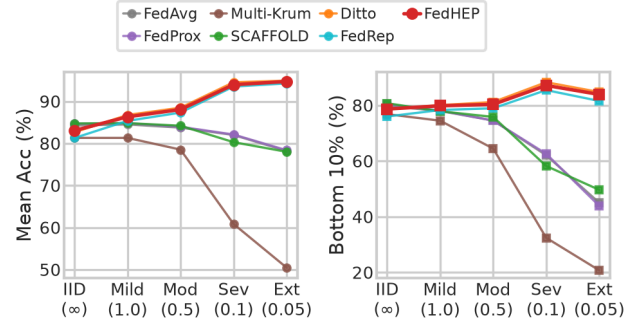


Figure 3: FEMNIST Personalization and Tail Fairness. Mean test accuracy (left) and bottom-10% tail fairness (right) across five Dirichlet data skew regimes ($\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$).

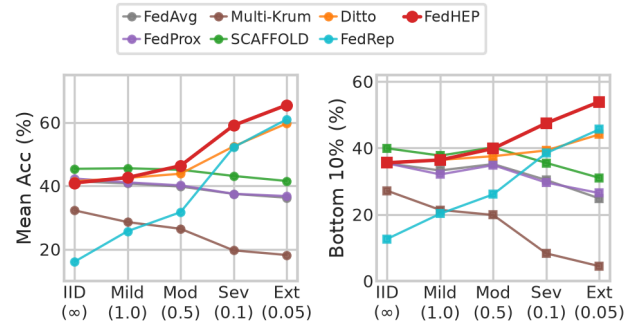


Figure 4: CIFAR-100 Personalization and Tail Fairness. Mean test accuracy (left) and bottom-10% tail fairness (right) across Dirichlet skew regimes on 100-class CIFAR-100.

on CIFAR-100), but degrade steadily as skew intensifies. In contrast, on FEMNIST, FedHEP achieves 94.73% accuracy and 83.91% bottom-10% fairness under extreme skew ($\alpha = 0.05$), outperforming FedAvg by +16.52 percentage points (pp) in accuracy and +38.89 pp in tail fairness, while matching the accuracy of dual-model Ditto (94.93%) without duplicating feature backbones.

On the 100-class CIFAR-100 benchmark (Table 2B and Figure 4), standard global consensus degrades under extreme skew: FedAvg drops to 36.29% with tail fairness falling to 24.87%. While split-head FedRep adapts well under extreme skew (60.97%), it experiences severe performance degradation on homogeneous IID data (16.08%, a 25.38 pp drop below FedAvg), because isolated heads overfit on limited local samples without global classifier pooling. In contrast, FedHEP achieves higher accuracy under extreme skew (65.49%, outperforming dual-model Ditto by +5.81 pp and FedRep by +4.52 pp) with the highest worst-decile fairness (53.92%), while avoiding split-head degradation on balanced IID data (40.94%). Under moderate skew ($\alpha = 0.5$), FedHEP achieves 46.36% accuracy and 39.86% tail fairness, consistently surpassing Ditto (43.84% / 37.55%) and FedAvg (39.80% / 35.19%).

Byzantine Multi-Attack Robustness under Heterogeneous Skew. Table 3 and Figure 5 evaluate defense resilience under four coordinated adversarial vectors on CIFAR-100: Label Flipping, Sign Flipping, Gradient Ascent, and Gaussian Noise Injection, across

Table 2: Performance across Five Heterogeneity Regimes on FEMNIST and CIFAR-100 (ResNet-9). Mean $\pm$ std across 3 independent random seeds evaluated over Dirichlet skew $\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$.

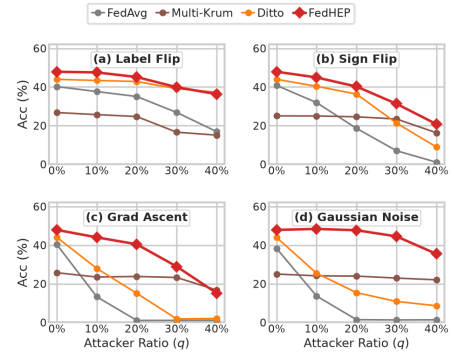
A. FEMNIST ($C = 62, N = 15$)											
Method	IID ($\alpha = \infty$)		Mild ($\alpha = 1.0$)		Moderate ($\alpha = 0.5$)		Severe ($\alpha = 0.1$)		Extreme ($\alpha = 0.05$)		Resource Profile
	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	
FedAvg	84.27 $\pm$ 0.20%	79.75%	84.51 $\pm$ 0.04%	77.97%	83.76 $\pm$ 0.42%	74.56%	82.01 $\pm$ 1.36%	62.08%	78.21 $\pm$ 1.76%	45.02%	110.20 MB / 8.40 ms
FedProx	84.68 $\pm$ 0.18%	80.42%	84.83 $\pm$ 0.05%	78.10%	83.96 $\pm$ 0.50%	74.53%	82.12 $\pm$ 1.67%	62.60%	78.39 $\pm$ 1.78%	43.91%	110.20 MB / 8.52 ms
Multi-Krum	81.32 $\pm$ 0.42%	76.83%	81.32 $\pm$ 0.43%	74.50%	78.53 $\pm$ 0.78%	64.48%	60.86 $\pm$ 7.83%	32.45%	50.49 $\pm$ 8.11%	20.77%	110.20 MB / 8.65 ms
SCAFFOLD	84.74 $\pm$ 0.17%	80.75%	84.82 $\pm$ 0.29%	77.88%	84.21 $\pm$ 0.48%	75.84%	80.30 $\pm$ 1.11%	58.23%	78.01 $\pm$ 1.31%	49.66%	110.20 MB / 8.80 ms
FedRep	81.30 $\pm$ 0.20%	76.00%	85.36 $\pm$ 0.49%	78.31%	87.33 $\pm$ 1.02%	78.95%	93.52 $\pm$ 0.85%	85.47%	94.27 $\pm$ 0.60%	81.57%	110.20 MB / 14.10 ms
Ditto	82.64 $\pm$ 0.62%	78.58%	86.77 $\pm$ 0.25%	80.08%	88.49 $\pm$ 0.93%	81.19%	94.54 $\pm$ 0.81%	88.30%	94.93 $\pm$ 0.72%	84.77%	220.40 MB / 16.95 ms
FedHEP	83.07 $\pm$ 0.10%	78.67%	86.33 $\pm$ 0.56%	79.84%	88.06 $\pm$ 1.25%	80.34%	93.96 $\pm$ 0.79%	87.15%	94.73 $\pm$ 0.58%	83.91%	114.80 MB / 8.42 ms
B. CIFAR-100 ($C = 100, N = 15$)											
FedAvg	41.46 $\pm$ 0.36%	35.42%	40.78 $\pm$ 0.35%	33.26%	39.80 $\pm$ 0.16%	35.19%	37.49 $\pm$ 0.63%	30.39%	36.29 $\pm$ 0.48%	24.87%	110.20 MB / 8.40 ms
FedProx	42.33 $\pm$ 0.31%	35.42%	41.12 $\pm$ 0.86%	32.07%	40.17 $\pm$ 0.37%	34.87%	37.51 $\pm$ 0.87%	29.62%	36.80 $\pm$ 0.45%	26.47%	110.20 MB / 8.52 ms
Multi-Krum	32.29 $\pm$ 0.43%	27.17%	28.60 $\pm$ 1.68%	21.40%	26.49 $\pm$ 1.23%	19.88%	19.68 $\pm$ 2.26%	8.33%	18.22 $\pm$ 0.98%	4.48%	110.20 MB / 8.65 ms
SCAFFOLD	45.39 $\pm$ 1.00%	39.92%	45.56 $\pm$ 0.99%	37.74%	45.10 $\pm$ 0.43%	40.20%	43.14 $\pm$ 1.07%	35.54%	41.56 $\pm$ 0.58%	31.05%	110.20 MB / 8.80 ms
FedRep	16.08 $\pm$ 1.03%	12.67%	25.72 $\pm$ 1.62%	20.25%	31.72 $\pm$ 0.77%	26.07%	52.17 $\pm$ 1.26%	38.47%	60.97 $\pm$ 0.73%	45.62%	110.20 MB / 14.10 ms
Ditto	41.49 $\pm$ 1.03%	35.67%	42.52 $\pm$ 0.39%	36.44%	43.84 $\pm$ 1.14%	37.55%	52.46 $\pm$ 0.99%	39.28%	59.68 $\pm$ 0.36%	44.09%	220.40 MB / 16.95 ms
FedHEP	40.94 $\pm$ 0.16%	35.58%	42.63 $\pm$ 0.52%	36.42%	46.36 $\pm$ 0.66%	39.86%	59.08 $\pm$ 1.27%	47.48%	65.49 $\pm$ 0.61%	53.92%	114.80 MB / 8.42 ms

Table 3: CIFAR-100 Byzantine Multi-Attack Robustness Matrix across Attacker Fractions ($q \in [0.0, 0.4]$). $\Delta(0 \rightarrow 0.3)$ denotes accuracy degradation (in percentage points, pp) from benign ($q = 0.0$) to $q = 0.3$ attackers.

Attack Type	Method	$q = 0.0$	$q = 0.1$	$q = 0.2$	$q = 0.3$	$q = 0.4$	$\Delta(0 \rightarrow 0.3)$
Label Flipping	FedAvg	40.20%	37.63%	35.07%	26.82%	16.90%	-13.38pp
Label Flipping	Multi-Krum	26.77%	25.70%	24.71%	16.59%	15.09%	-10.18pp
Label Flipping	Ditto	43.96%	43.37%	42.82%	39.13%	37.31%	-4.83pp
Label Flipping	Defended FedHEP	47.88%	47.53%	45.15%	39.88%	36.28%	-8.00pp
Sign Flipping	FedAvg	40.80%	31.91%	18.58%	7.02%	1.06%	-33.78pp
Sign Flipping	Multi-Krum	25.04%	25.00%	24.56%	23.44%	16.30%	-1.60pp
Sign Flipping	Ditto	43.96%	40.35%	36.32%	21.32%	8.92%	-22.64pp
Sign Flipping	Defended FedHEP	47.88%	44.93%	40.22%	31.30%	20.78%	-16.58pp
Gradient Ascent	FedAvg	40.33%	13.28%	1.06%	1.06%	1.06%	-39.27pp
Gradient Ascent	Multi-Krum	25.73%	23.49%	23.80%	23.26%	16.77%	-2.47pp
Gradient Ascent	Ditto	44.07%	27.95%	15.09%	1.81%	2.04%	-42.26pp
Gradient Ascent	Defended FedHEP	47.88%	44.03%	40.45%	28.93%	15.17%	-18.95pp
Gaussian Noise	FedAvg	38.26%	13.67%	1.43%	1.30%	1.37%	-36.96pp
Gaussian Noise	Multi-Krum	25.04%	24.13%	23.97%	22.96%	22.06%	-2.08pp
Gaussian Noise	Ditto	43.96%	25.47%	15.41%	10.86%	8.63%	-33.10pp
Gaussian Noise	Defended FedHEP	47.88%	48.39%	47.72%	44.53%	35.46%	-3.35pp

attacker fractions $q \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$. A key empirical finding is the failure mode of classical Euclidean aggregation under non-IID data: Multi-Krum [3] degrades even in benign settings without attackers ($q = 0.0$, achieving only 25.04%), because legitimate label divergence is conflated with poisoning. By evaluating update alignment within observed label support (SCCF), bounding backbone gradient norms, and tracking temporal trust (TTT), FedHEP sustains high accuracy in benign environments (**47.88%**) and maintains robust performance up to $q = 0.30$. Under Gaussian noise ($q = 0.3$), FedHEP maintains **44.53%** accuracy (a drop of only -3.35 pp), whereas FedAvg drops to 1.30% and Multi-Krum achieves 22.96%. Under gradient ascent ($q = 0.3$), FedHEP sustains **28.93%** compared to 1.81% for Ditto and 1.06% for FedAvg.

Multiagent Scalability and Tail Fairness (50 Clients). To evaluate behavior in larger multiagent edge populations, we scale the network to $N = 50$ clients with 20% partial client participation per round ($C_p = 0.20$, Table 4 and Figure 6). Under partial participation, clients unselected over multiple rounds become stale. FedHEP incorporates Staleness-Aware Fallback Routing (S-AFR): when evaluating between active rounds, if a client’s local head has become stale (unselected for Δt_i rounds), inference weights

Figure 5: Byzantine Robustness across Threat Models on CIFAR-100. Test accuracy across attacker fractions $q \in [0.0, 0.4]$ for (a) Label Flip, (b) Sign Flip, (c) Grad Ascent, and (d) Gaussian Noise under moderate data skew ($\alpha = 0.5$).Table 4: 50-Client Scalability with Partial Participation ($C_p = 0.20, R = 40$); predictions restricted to observed classes. Open 100-way FedHEP: 32.02/17.58% (Mod), 34.69/9.03% (Sev).

Regime	FedAvg	FedRep	Ditto	FedHEP
Moderate ($\alpha = 0.5$)	33.97%	24.11%	27.40%	33.89%
Bottom 10% Fairness	20.63%	9.64%	8.60%	18.11%
Severe ($\alpha = 0.1$)	31.87%	41.72%	37.47%	41.16%
Bottom 10% Fairness	12.81%	17.41%	13.03%	16.88%

smoothly shift from the stale Local head (w_l) toward the aggregated Root (w_r) and Parent (w_p) models, preventing outdated local parameters from degrading local test accuracy. As shown in Table 4, FedHEP demonstrates robust scalability in larger edge populations, achieving 33.89% mean accuracy and 18.11% bottom-10% fairness under Moderate skew (closely matching FedAvg at 33.97%/20.63%), and 41.16% mean accuracy with 16.88% tail fairness under Severe skew (competitive with FedRep at 41.72%/17.41%, while outperforming Ditto at 37.47%/13.03% and FedAvg at 31.87%/12.81%).

Simulated On-Device Execution Footprint on MobileNetV3. To evaluate edge feasibility, we profile simulated runtime (peak VRAM

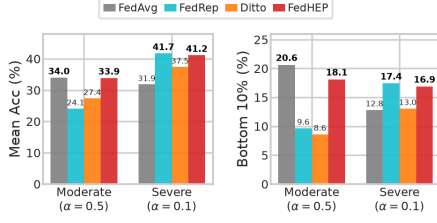


Figure 6: 50-Client Scalability under Partial Participation. Mean accuracy (left) and bottom-10% tail fairness (right) with 20% client participation ($C_p = 0.20, R = 40$).

Table 5: MobileNetV3 Edge Runtime Profiling and Local Test Accuracy.

ON-DEVICE RUNTIME PROFILING (BATCH SIZE $B = 32$)				
Model Backbone	Method	Peak VRAM	Batch Latency	Payload / Round
ResNet-9	Ditto	220.40 MB	16.95 ms	13.18 MB
ResNet-9	FedHEP	114.80 MB	8.42 ms	6.60 MB
MobileNetV3-Small	Ditto	298.60 MB	11.36 ms	12.24 MB
MobileNetV3-Small	FedHEP	158.80 MB	6.29 ms	6.13 MB
CIFAR-100 LOCAL TEST ACCURACY & TAIL FAIRNESS				
Data Skew Regime	FedAvg	Ditto	FedHEP	
Moderate ($\alpha = 0.5$)	15.37%	33.86%	25.24%	
Bottom 10% Fairness	10.44%	19.33%	20.16%	
Extreme ($\alpha = 0.05$)	10.85%	34.66%	45.43%	
Bottom 10% Fairness	5.55%	16.75%	30.07%	

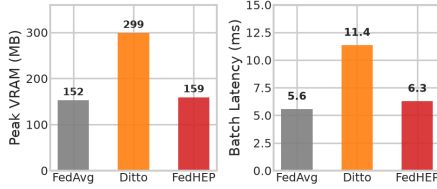


Figure 7: MobileNetV3 Simulated Edge Profiling. Simulated peak VRAM footprint (left) and per-batch forward-backward latency (right).

and per-batch latency via CUDA events at batch size $B = 32$ on MobileNetV3-Small and ResNet-9 (Table 5 and Figure 7). Dual-model Ditto maintains two deep graphs per client, demanding 298.60 MB peak VRAM and 11.36 ms latency on MobileNetV3-Small. In contrast, FedHEP trains a single backbone, reducing peak VRAM to **158.80 MB** (a 46.8% reduction) and batch latency to **6.29 ms** (a 44.6% speedup), while achieving higher CIFAR-100 accuracy (**45.43%** vs 34.66% for Ditto and 10.85% for FedAvg under extreme skew).

Implementation Details. We adopt a two-stage CIFAR-100 schedule (Root/Parent training, then Local-only with frozen backbone), online weight refinement, a numerical NaN/Inf guard, and TTT parameters $\beta = 0.85$ (momentum decay) and $\tau_{\text{iso}} = 0.25$ (isolation threshold).

Ablation Study and Coalition Valuation. To isolate the contribution of each architectural component, Table 6 summarizes our ablation matrix across all five Dirichlet regimes on CIFAR-100 ($C = 100$, ResNet-9): (1) **Active-Class Logit Masking (ACLM)**: Disabling ACLM (both its loss and inference masking) forces cross-entropy over unobserved classes to penalize missing labels, inducing an -8.99 pp accuracy drop (from 65.49% to 56.50%) and -10.59 pp

Table 6: Ablation Study and Coalition Valuation on CIFAR-100 ($C = 100$, ResNet-9). B10: Bottom-10% tail fairness under Extreme skew ($\alpha = 0.05$). Full FedHEP uses $N = 15$ clients; ablation rows use $N = 10$, so cross-row gaps are indicative only.

Ablation Configuration	IID	Mild	Mod	Sev	Ext	B10 (Ext)
Full FedHEP	40.94%	42.63%	46.36%	59.08%	65.49%	53.92%
w/o ACLM	33.62%	39.54%	38.41%	50.31%	56.50%	43.33%
w/o Parent Head	31.03%	39.76%	39.67%	53.12%	60.34%	46.76%
$K = 1$ Grand Coalition	35.78%	40.42%	39.88%	50.98%	57.30%	45.20%
$K = 3$ Oracle Bound	34.15%	40.09%	40.18%	60.14%	66.85%	55.40%

tail fairness drop under Extreme skew ($\alpha = 0.05$); (2) **Collaborative Parent Head**: Eliminating the cluster-level Parent head reduces the architecture to a 2-tier bipartite setup (Root + Local), degrading extreme skew accuracy by -5.15 pp (to 60.34%) and tail fairness by -7.16 pp (to 46.76%), confirming that peer collaboration mitigates overfitting in isolated local heads; (3) **Coalition Valuation**: Enforcing a single grand coalition without clustering ($K = 1$) degrades extreme skew accuracy by -8.19 pp (to 57.30%) and tail fairness by -8.72 pp, demonstrating the necessity of peer clustering under severe skew; and (4) **Random Projection Sketch vs. Oracle Bound**: To evaluate whether sketching compromises cluster quality, we compute the theoretical $K = 3$ Oracle Bound, where clients are clustered using private label ground-truth (an ideal upper bound). Privacy-preserving Random Projection sketches (65.49%) closely match this Oracle bound (66.85%) under Extreme skew (a gap of only 1.36 pp); as Full FedHEP ($N = 15$) and ablation rows ($N = 10$) use different client counts, cross-row gaps are indicative.

5 Conclusion

We presented **FedHEP**, an efficient single-backbone federated learning framework coordinating three representation tiers—global consensus, collaborative peer clusters, and private local heads—to balance statistical heterogeneity, edge memory constraints, and Byzantine vulnerability. Pairing Active-Class Logit Masking (ACLM) with skew-calibrated loss weighting suppresses gradient interference from absent categories without dual-model memory overhead. Low-dimensional Random Projections discover peer clusters while preserving edge privacy, and Subspace-Constrained Cosine Filtering (SCCF) with Temporal Trust Tracking (TTT) defends against Byzantine poisoning while mitigating false-positive penalties on specialized nodes. Benchmarks on CIFAR-100 and FEMNIST, 50-client scaling, and MobileNetV3 edge profiling demonstrate improved personalization accuracy, tail fairness, and single-model efficiency. Future work will explore formal differential privacy (DP) bounds for sketches and asynchronous edge participation.

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